

Homework-1

Singer classification

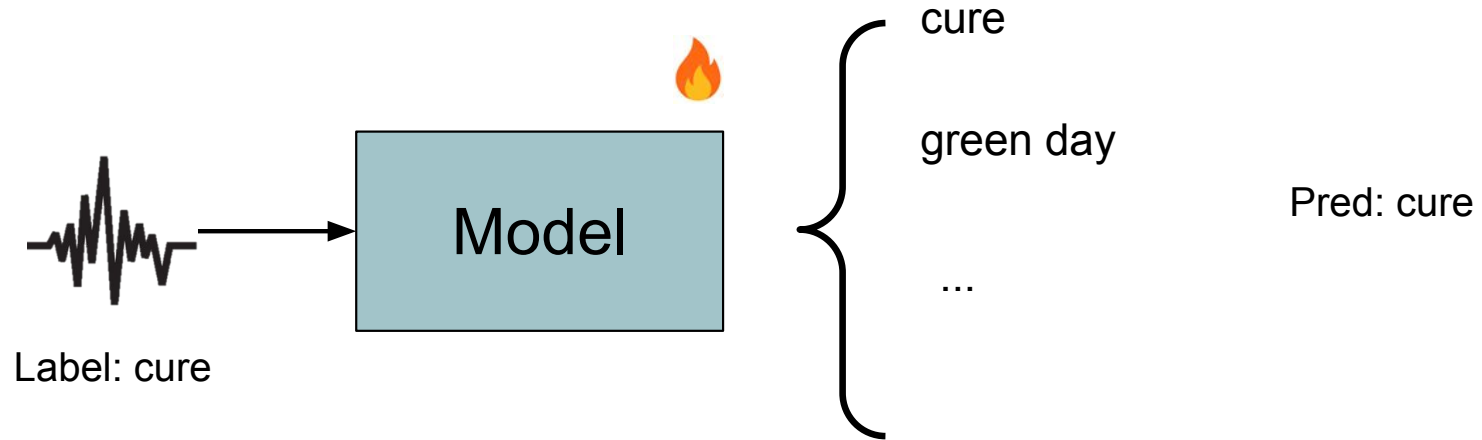
HW1 TA: 李維釗 (Wei-Jaw Lee)

Office hour: Mon. 13:30~15:00 @BL505

Outline

- Overview
- Detailed Explanation
- Submission
- Scoring
- Rules
- Timeline

Overview: Single-label Classification

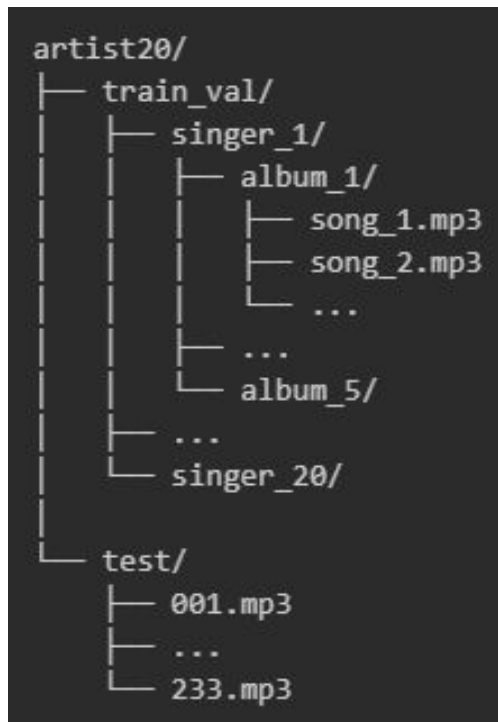


Detailed Explanation

- Dataset
- Tasks
 - a. Train a Traditional Machine Learning Model
 - b. Train a Deep Learning Model
- Evaluation
- Optional

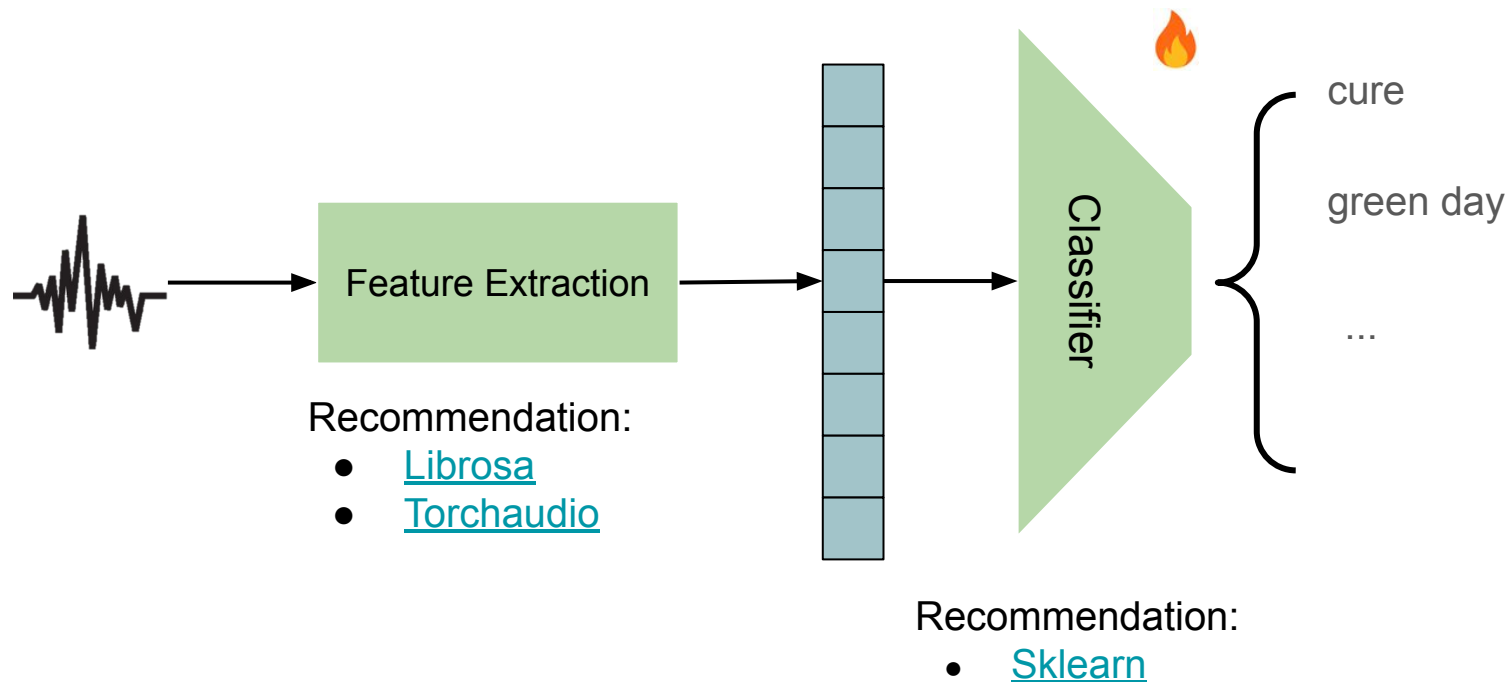
Dataset: Artist20

- Source: [Link](#)
- 20 artists, 120 albums, 1413 tracks
- Train / Validation / Test
Album-level split^[1]: 4 / 1 / 1
Tracks number: 949 / 231 / 233
- dataset size: 1.28G
Sample rate: 16000Hz
Ext: mp3
Channel: Mono
Length: full song



[1]: Hsieh et al, "Addressing the confounds of accompaniments in singer identification," ICASSP 2020

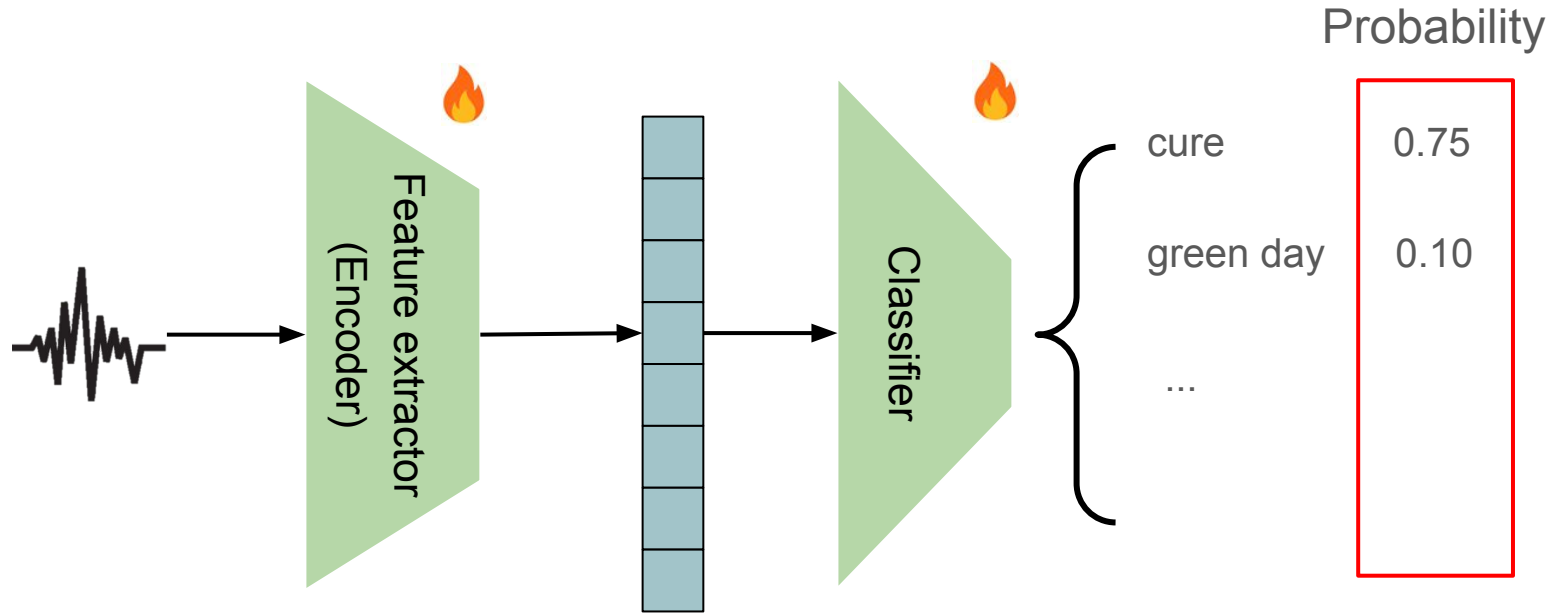
Task 1: Train a Traditional Machine Learning Model



Task 1: Train a Traditional Machine Learning Model

- Train a traditional ML model (e.g. k-NN, SVM, random forest) with any features extracted from the audio
- Need to report **the features** you use and the **model implementation** clearly
- Need to report the **validation** result with **confusion matrix**, **top1 accuracy**, and **top3 accuracy**
- Remember to utilize standardization (e.g. mean, std), pooling and normalization to ensure consistent feature scales, reducing overfitting, and improving model stability and performance during training

Task 2: Train a Deep Learning Model



Task 2: Train a Deep Learning Model

Some reference methods

- music classification: [Ref](#)
- singer recognition:
 - 19[ijcnn] [Musical artist classification with convolutional recurrent neural networks](#)
 - 20[icassp] [Addressing the confounds of accompaniments in singer identification](#)
 - 21[aaai] [Positions, channels, and layers fully generalized non-local network for singer](#)
 - 23[ismir] [Singer identity representation learning using self-supervised techniques](#)
- speaker recognition and verification
 - 22[icassp] [Domain Adaptation for Speaker Recognition in Singing and Spoken Voice](#)

Task 2: Train a Deep Learning Model

- Train a deep learning model from scratch
- Need to **report how to implement** the model clearly
- If you referred to some previous work, **cite the paper or code appropriately**
- Need to **report the validation result** with confusion matrix, top1 accuracy, and top3 accuracy
- Need to **submit the testing set result** to NTUCool

Task 2: Train a Deep Learning Model

Some optional baselines (not required)

- Use pre-trained models
 - 23[apsipa] [Toward leveraging pre-trained self-supervised frontends for automatic singing voice understanding tasks- three case studies](#)
 - 23[ismir] [On the effectiveness of speech self-supervised learning for music](#)
- Audio Language Model
 - 25[arxiv] [Audio Flamingo 3: Advancing Audio Intelligence with Fully Open Large Audio Language Models](#)
 - [Qwen](#)

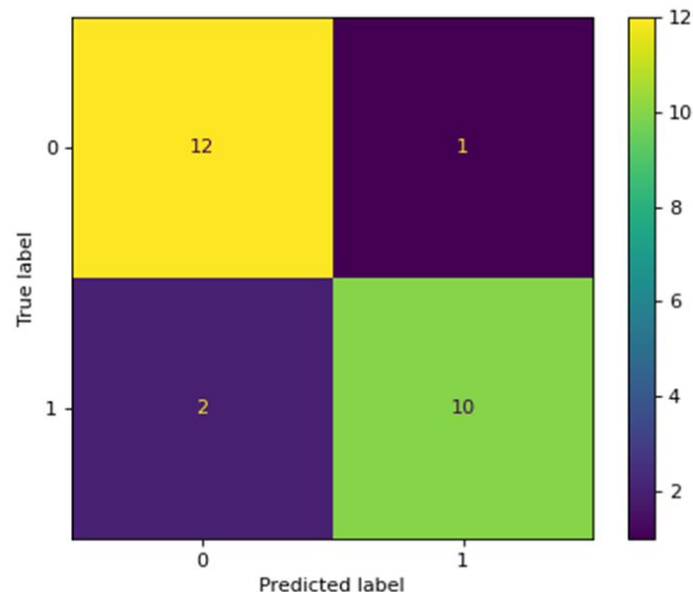
Evaluation

1. Confusion matrix

A table that summarizes the performance of a classification algorithm by comparing the predicted labels with the actual labels

2. Top k accuracy (Top 1 and Top 3)

For a given k, if the correct label is within the top k predicted classes, it's counted as correct; otherwise, it's counted as incorrect



Only use the test set for evaluation; don't use it at all for training

Data Pre-processing / Data Augmentation

Augmentation

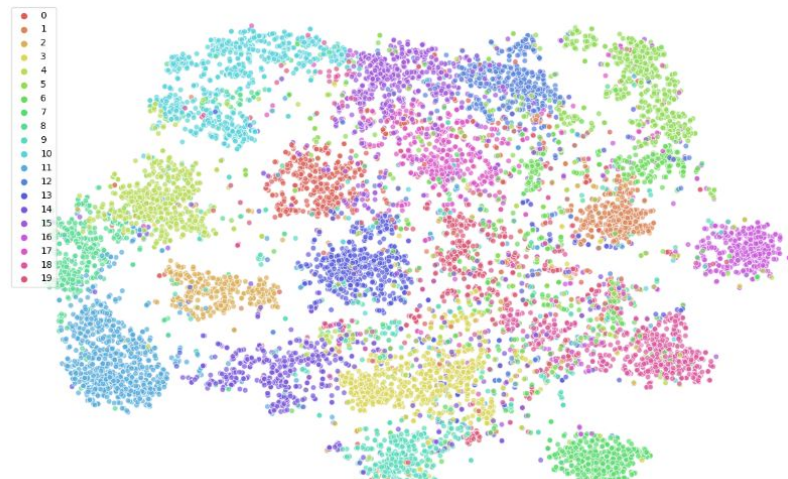
- [torchaudio](#)

Source separation

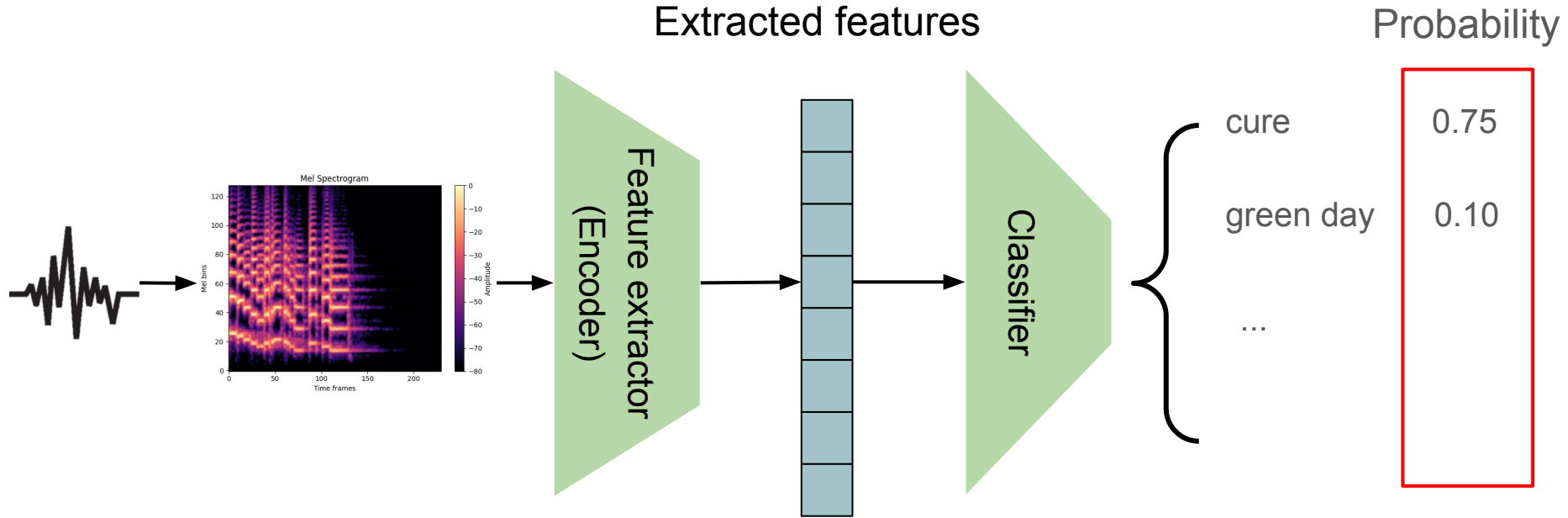
- open-unmix
- spleeter
- demucs

Optional

- Record your own singing voice or use a recording of a singer you like as the input to the best model you train; show the mel-spectrograms and discuss the result
- Use [t-SNE](#) to visualize the learned embeddings; use colors to mark different singers



Optional (cont.)



Submission

- Report (to NTU cool)
- Prediction (to NTU cool)
- Code & model checkpoint for inference (to your cloud drive)

We will randomly select several classmates' code to run inference on your model and run the score on your results, so please ensure that the files you upload include trained model which can successfully execute the entire inference process and the generation results

Don't upload: training data, testing data, preprocessed data, others model, cache file

Report

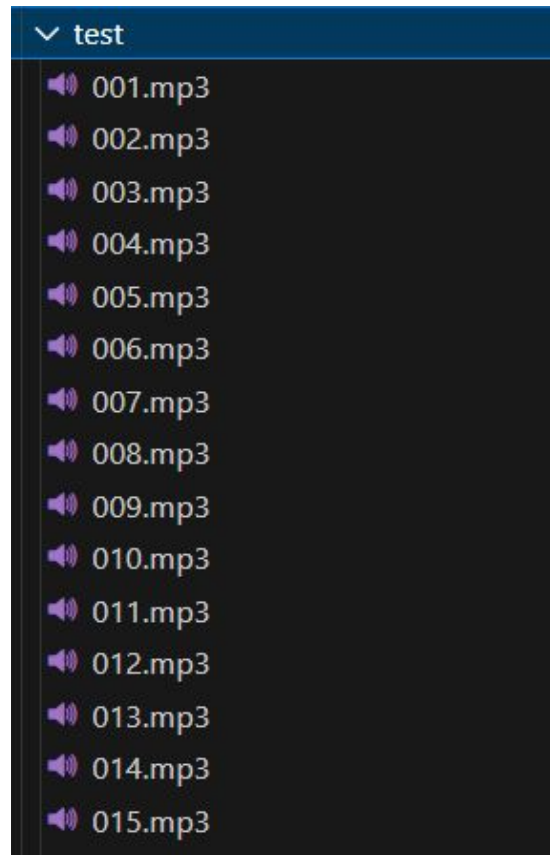
- Write with PPT or PPT-like format (in 16:9 aspect ratio)
- Upload **studentID_report.pdf** (ex: r12345678_report.pdf)
- Please create a report that is clear and can be understood without the need for oral explanations
- There is **no specific length requirement**, but it should clearly communicate the experiments conducted and their results; approximately 10 pages is a suggested standard, but not a strict limitation

Prediction

- 233 tracks testing audio
- File name: id.mp3
- ID ranging from 001 to 233
- results format:

student_ID.json file:

```
{  
    'id': [top1, top2, top3],  
    ...  
}
```

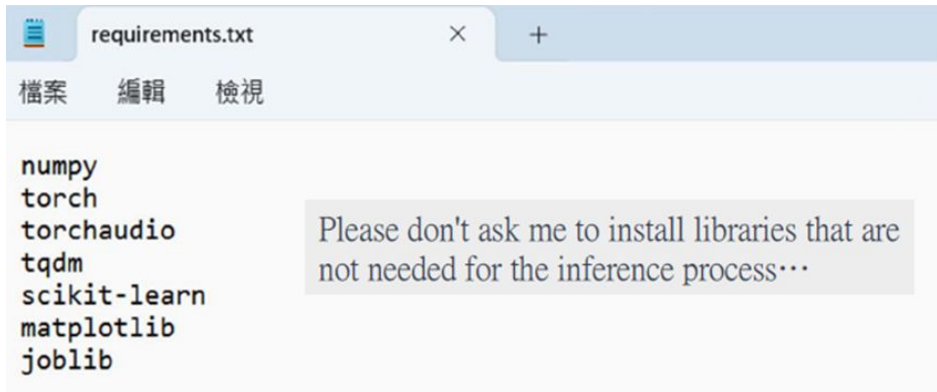


Code & model checkpoint

- Upload all your source code and model to a cloud drive, open access permissions, and then upload the link to the NTU Cool assignment HW0_report in comments, as well as include it on the first page of the report
- You will need to upload requirements.txt
- I'll run :

```
pip install -r requirements.txt
```

If you have used third-party programs that cannot be installed directly via 'pip install,' please write the URL and install method command by command on **your readme file**.



Code

- You will also need to upload [readme file](#) to guide me on how to perform inference on your model. (I'll use the same test set, ensure I can run your code with the test set path in my device.)
- The inference code should output the **top 3 predictions in descend accuracy order**

Scoring

- HW1 accounts for 20% of the total grade
 - Report: 50%
 - Prediction: 50%
 - $\text{top 1 accuracy} + 0.5 * \text{top 3 accuracy}$

Rules

- Don't cheat
- Don't use extra data
- Don't use the test data while training or tuning your model
- Can use public codes by properly citing in your report

Timeline

- W2 – 09/11 (Thursday): Announcement of HW1
- W6 – 10/08 (Wednesday 23:59pm): Deadline

Late submission: 1 day (-20%), 2 days (-40%), 3 days (-60%), after that (no score)

ALL things you need to do before 10/08 23:59

- StudentID_report.pdf
- Cloud drive link
 - readme file
 - Requirements.txt
 - Codes and model to run inference
 - Others codes
- Test set prediction

